

Overview of AlphaGo

Subject: AlphaGo

1.

Go is considered much more difficult for computers to win than other games such as chess, because its strategic and aesthetic nature makes it hard to directly construct an evaluation function, and its much larger branching factor makes it prohibitively difficult to use traditional AI methods such as alpha–beta pruning, tree traversal and heuristic search. Almost two decades after IBM's computer Deep Blue beat world chess champion Garry Kasparov in the 1997 match, the strongest Go programs using artificial intelligence techniques only reached about amateur 5-dan level, and still could not beat a professional Go player without a handicap. In 2012, the software program Zen, running on a four PC cluster, beat Masaki Takemiya (9p) twice at five- and four-stone handicaps. In 2013, Crazy Stone beat Yoshio Ishida (9p) at a four-stone handicap. According to DeepMind's David Silver, the AlphaGo research project was formed around 2014 to test how well a neural network using deep learning can compete at Go. AlphaGo represents a significant improvement over previous Go programs. In 500 games against other available Go programs, including Crazy Stone and Zen, AlphaGo running on a single computer won all but one. In a similar matchup, AlphaGo running on multiple computers won all 500 games played against other Go programs, and 77% of games played against AlphaGo running on a single computer. The distributed version in October 2015 was using 1,202 CPUs and 176 GPUs.

2.

Similar systems Facebook has also been working on its own Go-playing system darkforest, also based on combining machine learning and Monte Carlo tree search. Although a strong player against other computer Go programs, as of early 2016, it had not yet defeated a professional human player. Darkforest has lost to CrazyStone and Zen and is estimated to be of similar strength to them. DeepZenGo, a system developed with support from video-sharing website Dwango and the University of Tokyo, lost 2–1 in November 2016 to Go master Cho Chikun, who holds the record for the largest number of Go title wins in Japan. A 2018 paper in Nature cited AlphaGo's approach as the basis for a new means of computing potential pharmaceutical drug molecules. Systems consisting of Monte Carlo tree search guided by neural networks have since been explored for a wide array of applications.

3.

In the Future of Go Summit held in Wuzhen in May 2017, AlphaGo Master played three games with Ke Jie, the world No.1 ranked player, as well as two games with several top Chinese professionals, one pair Go game and one against a collaborating team of five human players. Google DeepMind offered 1.5 million dollar winner prizes for the three-game match between Ke Jie and Master while the losing side took 300,000 dollars. Master won all three games against Ke Jie, after which AlphaGo was awarded professional 9-dan by the Chinese Weiqi Association. After winning its three-game match against Ke Jie, the top-rated world Go player, AlphaGo retired. DeepMind also disbanded the team that worked on the game to focus on AI research in other areas. After the Summit, DeepMind published 50 full length AlphaGo vs AlphaGo matches, as a gift to the Go community.

4.

Impacts on Go The documentary film AlphaGo raised hopes that Lee Sedol and Fan Hui would have benefitted from their experience of playing AlphaGo, but as of May 2018, their ratings were little changed; Lee Sedol was ranked 11th in the world, and Fan Hui 545th. On 19 November 2019, Lee announced his retirement from professional play, arguing that he could never be the top overall player of Go due to the increasing dominance of AI. Lee referred to them as being "an entity that cannot be defeated".

5.

Go community Go is a popular game in China, Japan and Korea, and the 2016 matches were watched by perhaps a hundred million people worldwide. Many top Go players characterized AlphaGo's unorthodox plays as seemingly-questionable moves that initially befuddled onlookers, but made sense in hindsight: "All but the very best Go players craft their style by imitating top players. AlphaGo seems to have totally original moves it creates itself." AlphaGo appeared to have unexpectedly become much stronger, even when compared with its October 2015 match where a computer had beaten a Go professional for the first time ever without the advantage of a handicap. The day after Lee's first defeat, Jeong Ahram, the lead Go correspondent for one of South Korea's biggest daily newspapers, said "Last night was very gloomy... Many people drank alcohol." The Korea Baduk Association, the organization that oversees Go professionals in South Korea, awarded AlphaGo an honorary 9-dan title for exhibiting creative skills and pushing forward the game's progress. China's Ke Jie, an 18-year-old generally recognized as the world's best Go player at the time, initially claimed that he would be able to beat AlphaGo, but declined to play against it for fear that it would "copy my style". As the matches progressed, Ke Jie went back and forth, stating that "it is highly likely that I (could) lose" after analysing the first three matches, but regaining confidence after AlphaGo displayed flaws in the fourth match. Toby Manning, the referee of AlphaGo's match against Fan Hui, and Hajin Lee, secretary general of the International Go Federation, both reason that in the future, Go players will get help from computers to learn what they have done wrong in games and improve their skills. After game two, Lee said he felt "speechless": "From the very beginning of the match, I could never manage an upper hand for one single move. It was AlphaGo's total victory." Lee apologized for his losses, stating after game three that "I misjudged the capabilities of AlphaGo and felt powerless." He emphasized that the defeat was "Lee Se-dol's defeat" and "not a defeat of mankind". Lee said his eventual loss to a machine was "inevitable" but stated that "robots will never understand the beauty of the game the same way that we humans do." Lee called his game four victory a "priceless win that I (would) not exchange for anything."